Google Data Analytics Capstone Project Report

Cyclistic Bike-Share Usage Analysis | October 2025–September 2026 | Prepared by: technikhi

1. Business Background and Objective

Business background

Cyclistic is a bike-share company operating in Chicago. It offers single-ride and full-day passes for casual riders, as well as annual memberships. Lily Moreno, Director of Marketing, believes that increasing annual memberships is important to the company's future growth.

Business task

This analysis compares how casual riders and annual members use Cyclistic bikes. The aim is to identify differences in ride volume, duration, timing, bike type and starting-station usage that could inform strategies to convert casual riders into annual members.

Stakeholders

- Lily Moreno, Director of Marketing
- Cyclistic Marketing Analytics Team
- Cyclistic Executive Team

Working hypothesis

Casual riders already use Cyclistic's service, but their usage patterns may differ from those of annual members. Understanding these differences may help the marketing team develop more relevant membership campaigns.

2. Data Sources and Quality

Source and reporting period

The analysis uses 12 months of trip data, from October 2025 to September 2026. The data was published by Motivate International Inc. under its Data License Agreement.

Source: **Divvy Trip Data** — https://divvy-tripdata.s3.amazonaws.com/index.html

The dataset excludes personally identifiable information. Trips therefore cannot be linked to individual riders or payment details. For example, the analysis cannot determine whether a casual rider lives within the service area or has purchased multiple single-ride or day passes.

ROCCC assessment

Criterion	Assessment
Reliable	The data is published through the company's data source, although each trip record cannot be independently verified.
Original	The dataset is provided by Motivate International Inc., the publisher.

Comprehensive	The fields support comparisons of ride frequency, duration, rider category and bike type. Missing station information limits some station-level analysis.
Current	The selected 12-month period is relevant to this analysis, but the records are historical rather than real-time.
Cited	The data source and licensing information are publicly documented.

Data integrity notes

The dataset contains the key fields required for this analysis, including ride IDs, rider categories, bike types, and trip start and end timestamps. These fields provide the information needed to calculate trip durations and compare usage patterns between casual riders and annual members.

Data integrity checks included examining the uniqueness of ride IDs and reviewing the data for inconsistencies during the cleaning process. The check identified 19 duplicate ride_id entries. These records were retained because they did not materially affect the overall analysis. Their presence is acknowledged as a data-quality limitation.

During data preparation, the January 2026 .zip file was found to have been incorrectly labelled as 202501 after extraction. The filename was corrected to 202601 to reflect the actual reporting month before the data was processed.

3. Data Cleaning and Preparation

The 12 monthly CSV files were imported into Python using the Pandas library. Each monthly dataset was cleaned and prepared individually before being combined into a single master dataset for analysis.

The following data preparation steps were performed:

1. **Data cleaning and validation:** Each monthly dataset was reviewed and cleaned before consolidation to ensure consistent data formats and categorisation.
2. **Removal of unnecessary columns:** Latitude and longitude columns were removed as they were not required for the main analysis.
3. **Date and time conversion:** Trip start and end timestamps were converted into datetime formats to enable date-based calculations and analysis.
4. **Ride duration calculation:** A new ride_length field was created by calculating the difference between the trip start and end timestamps. The resulting duration was expressed in minutes.
5. **Feature extraction:** The day of the week was extracted from the trip start timestamp to support analysis of usage patterns across different days.
6. **Removal of invalid trip durations:** Records with negative or zero trip durations were removed, as were trips lasting less than 60 seconds or longer than 24 hours. These thresholds were applied to exclude records that could represent invalid or atypical trips.
7. **Handling missing station information:** Rows containing missing values in the station name fields were removed from the dataset before analysis. Consequently, these records were excluded from all subsequent calculations and visualisations, including overall ride counts and duration comparisons.
8. **Standardisation:** Rider categories (member and casual) and bike-type labels were standardised to ensure consistent categorisation during analysis.

9. **Duplicate check:** Each monthly dataset was checked for duplicate ride IDs. Nineteen duplicate entries were identified across the combined dataset and retained, as they did not materially affect the overall analysis.
10. **Data consolidation:** After the individual monthly datasets had been cleaned and prepared, they were combined into a single master dataset covering the reporting period.

    Following these steps, the resulting master dataset was used for the subsequent analysis and visualisation.

4. Analysis and Findings

All ride counts and percentages presented in this section are calculated from the cleaned dataset after the exclusions described in Section 3.

4.1 Ride volume and seasonality

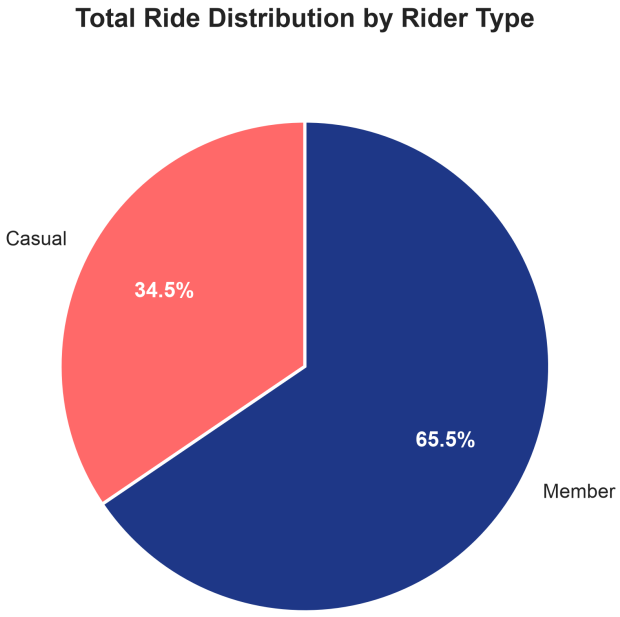


Figure 1: Total Ride Distribution by Rider Type

Across the reporting period, the dataset contains 1,396,474 casual rides and 2,652,971 member rides. Members recorded approximately 1.9 times as many rides as casual riders and accounted for about 65.5% of all rides.

- Members recorded more rides than casual riders in every month.
- Ride volumes for both groups were lower during winter and increased from spring into summer.
- Casual ride volume peaked in July at 224,405 rides; member ride volume peaked in August at 337,443 rides.

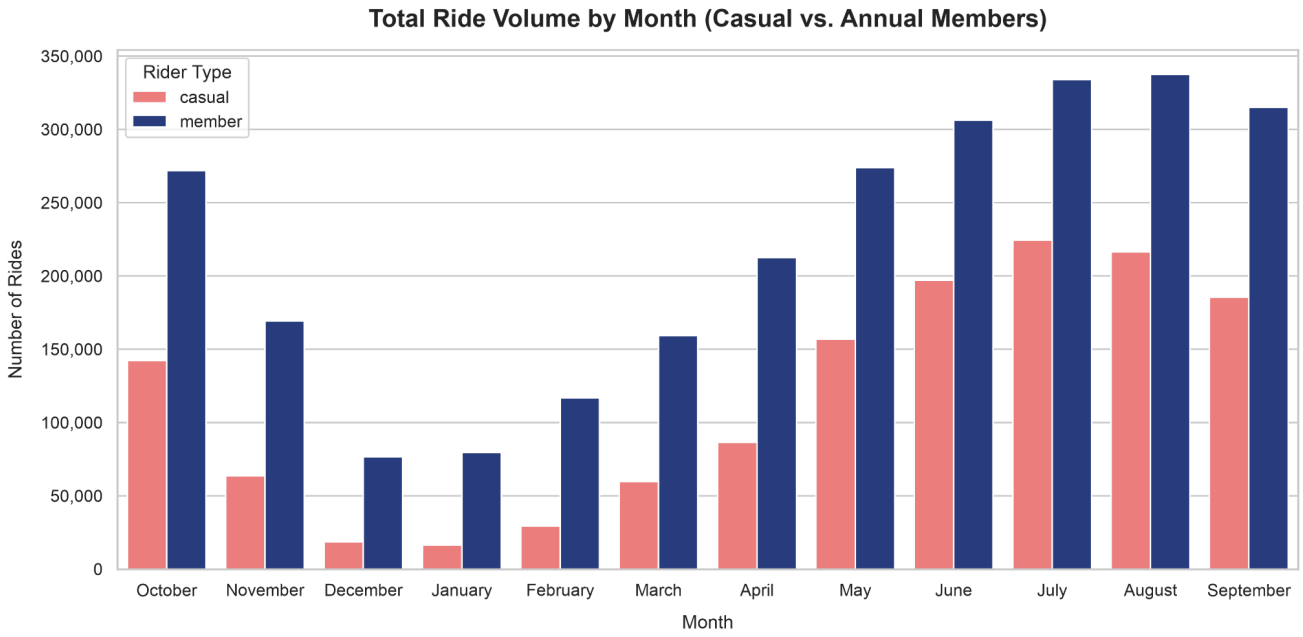


Figure 2: Monthly Ride Volume by Rider Category

4.2 Ride duration

Metric	Casual riders	Annual members
Mean duration	20.30 minutes	12.40 minutes
Median duration	11.97 minutes	8.70 minutes

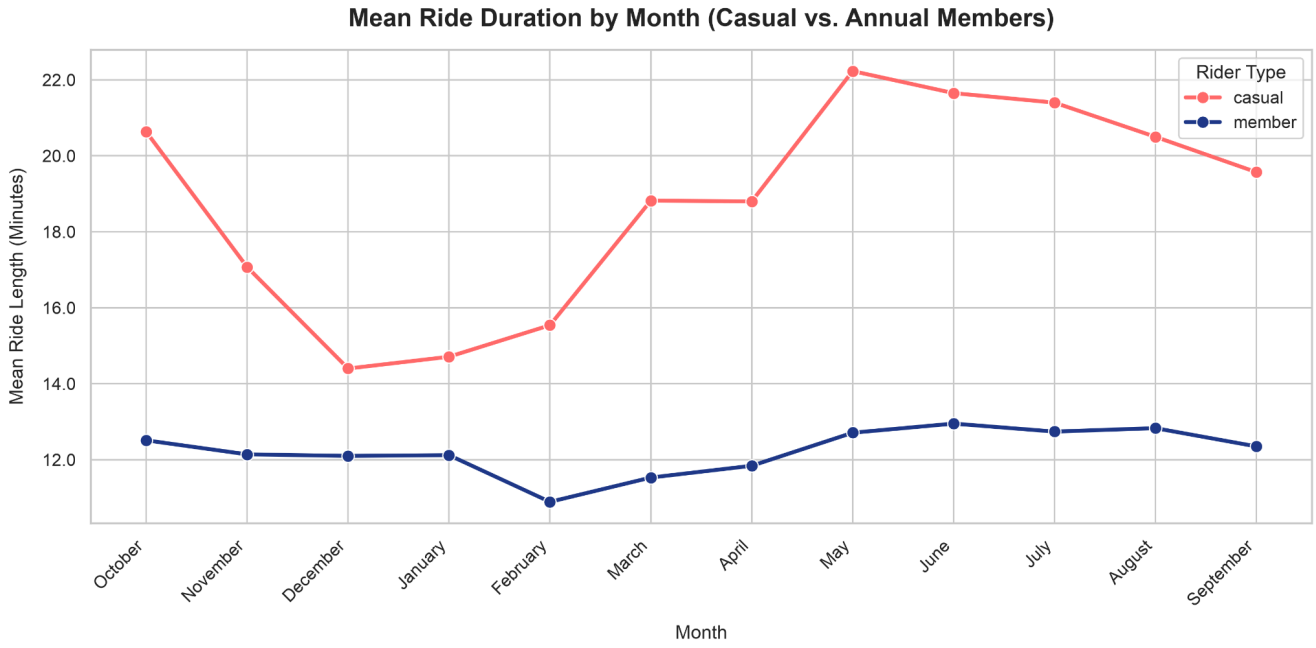


Figure 3: Average Monthly Ride Duration by Rider Category

Casual rides were approximately 64% longer on average than member rides. The median shows the same pattern, so the difference is not explained solely by a small number of unusually long rides.

- Casual riders' mean ride duration was higher every month.
- Casual mean duration peaked in May at 22.23 minutes.
- Member mean duration was comparatively stable, ranging from 10.89 to 12.95 minutes.

4.3 Weekday and weekend patterns

Rider category	Weekday rides	Weekend rides	Weekend share
Casual	876,110	520,364	37.26%
Member	2,054,315	598,656	22.57%

Weekend rides represented a larger share of casual rides than member rides. The casual weekend share was approximately 1.65 times the member share, and casual riders had a higher weekend share in every month. The casual share peaked at 43.31% in May; the member share also peaked in May, at 27.53%. Both groups had lower weekend shares from December through February.

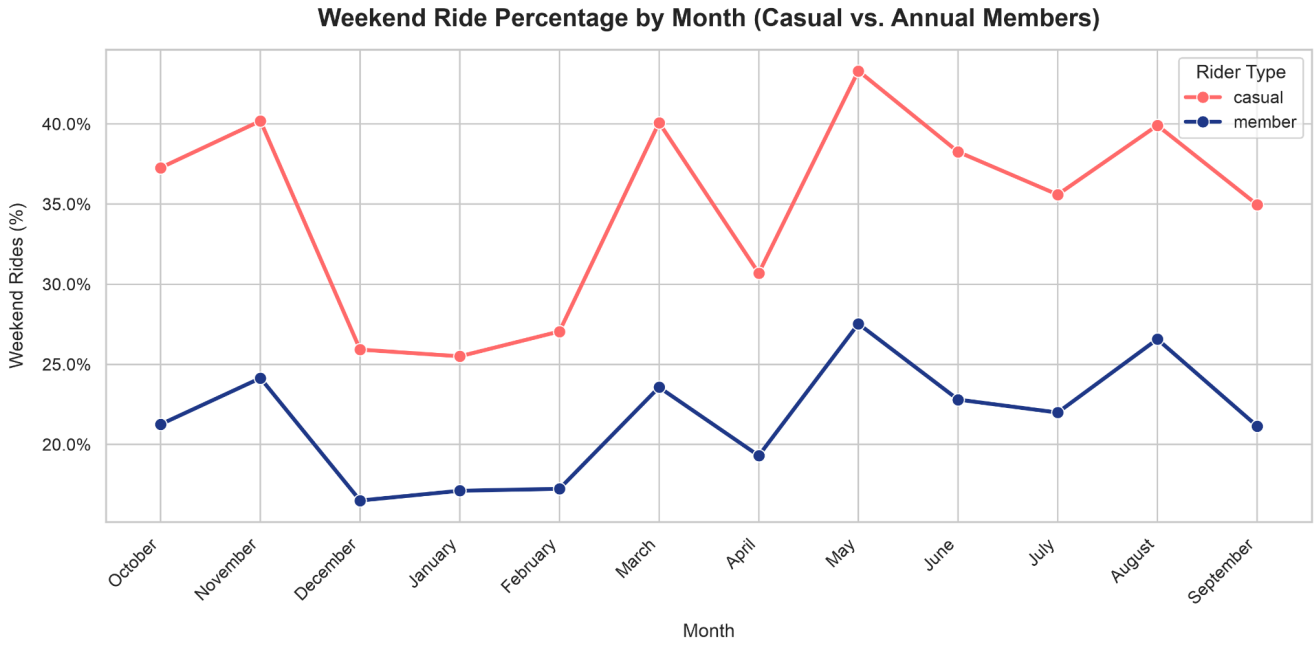


Figure 4: Monthly Weekend Ride Share by Rider Category

Rider category	Weekday mean duration	Weekend mean duration
Casual	18.52 minutes	23.31 minutes
Member	12.01 minutes	13.76 minutes

Casual riders' mean ride duration increased by approximately 26% on weekends, compared with about 15% for members. Together, the ride-share and duration results suggest that casual usage is more weekend-oriented, while member usage is more concentrated on weekdays.

Saturday was the most common day for casual riders in 9 of the 12 months. For members, a weekday was the most common day in all 12 months. These modal-day results identify the most frequent day in each group, but do not indicate how large the difference was compared with other days.

4.4 Starting-station patterns

Navy Pier was the most popular starting station overall, accounting for 58,932 rides, followed by DuSable Lake Shore Dr & Monroe St with 33,174 rides.

A comparison of the most frequently used starting stations by rider category revealed differences in station usage. Among casual riders, Navy Pier ranked first with 44,897 rides. Other frequently used stations included DuSable Lake Shore Dr & Monroe St, Michigan Ave & Oak St, Millennium Park, Shedd Aquarium, Theater on the Lake, DuSable Harbor, Adler Planetarium and Field Museum. Many of these stations are located near prominent lakefront, recreational and visitor destinations.

In contrast, members' most frequently used starting stations included Canal St & Madison St, State St & Chicago Ave, Clinton St & Jackson Blvd, and several locations along Wells Street. These stations are primarily situated in central urban areas.

The findings suggest that casual riders make greater use of stations near recreational and visitor destinations, while members show greater use of stations in central urban areas. However, station-level ride counts alone cannot establish the purpose of individual trips or whether riders are tourists, commuters or local residents.

4.5 Bike-type usage

Bike type	Casual rides	Member rides
Electric	61.03%	53.47%
Classic	38.97%	46.53%

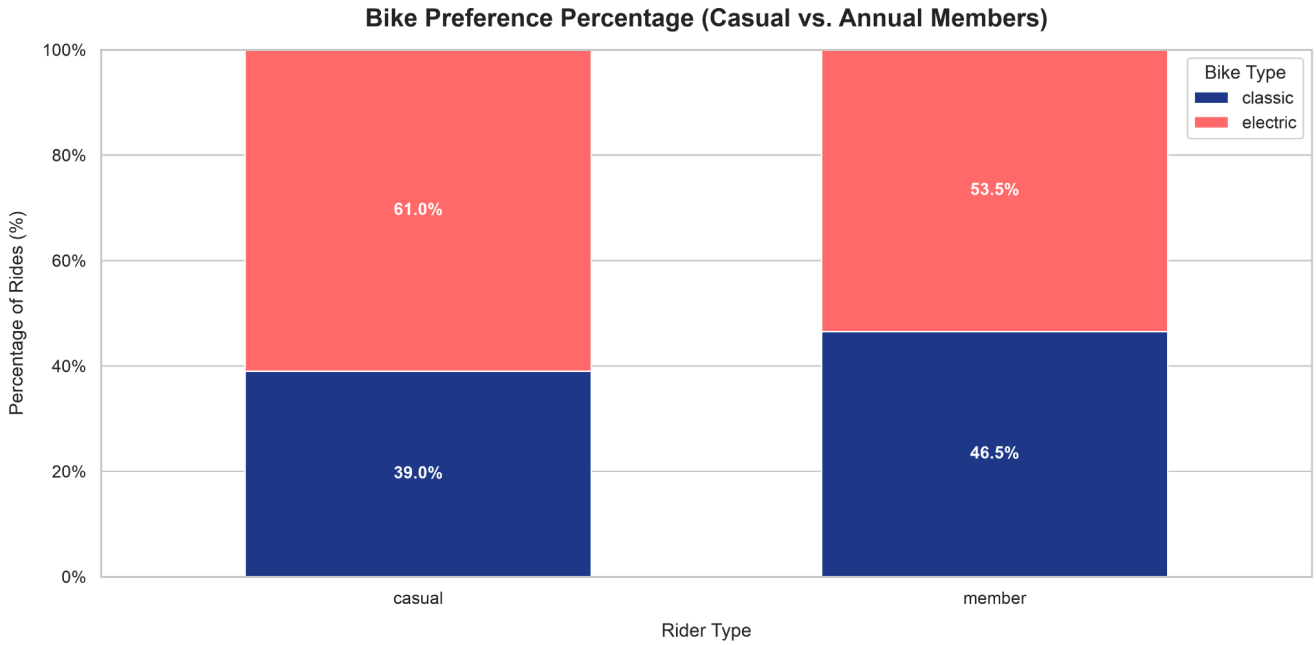


Figure 5: Ride Distribution by Bike Type and Rider Category

Electric bikes were used more often than classic bikes by both groups. The electric-bike share was 7.56 percentage points higher among casual rides than member rides. This makes bike type a potentially relevant factor for membership marketing, but these figures describe rides, not unique customers' preferences.

4.6 Ride start times and peak hours

Member ride starts show a morning peak at 8 AM (197,617 rides) and a larger evening peak at 5 PM (290,494 rides). Casual ride counts rise more gradually through the morning and afternoon, peaking at 5 PM (133,298 rides), without a comparable morning surge.

The member morning-and-evening pattern is consistent with commuting-related use, while the casual pattern may reflect more flexible ride timing. These are interpretations rather than confirmed trip purposes. Because the hourly table combines all days of the week, the morning member peak may be driven disproportionately by weekdays.

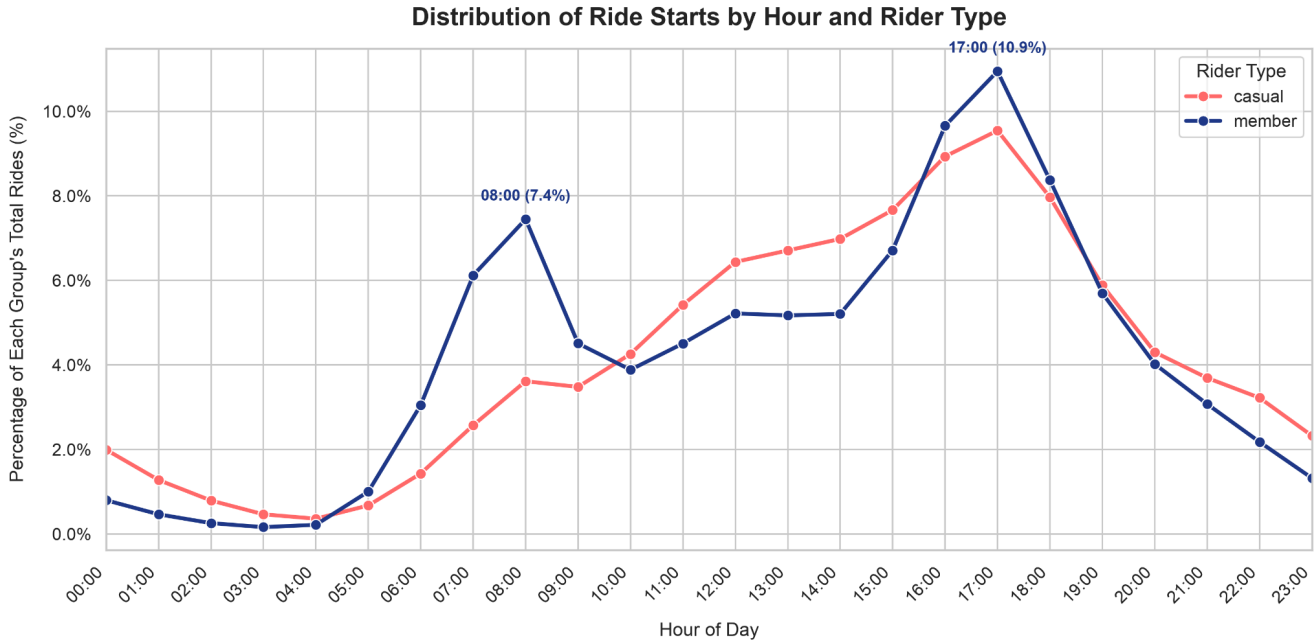


Figure 6: Hourly Ride Distribution by Rider Category

5. Key Takeaways

- Members generate more rides overall, accounting for approximately 65.5% of recorded rides.
- Casual rides are longer: mean duration is 20.30 minutes compared with 12.40 minutes for members.
- Casual riders have a higher weekend ride share (37.26% versus 22.57%) and their rides are longer on weekends.
- Usage patterns vary by starting station and bike type. Casual riders make greater use of several stations near lakefront and visitor destinations and have a higher share of electric-bike rides than annual members.
- Member ride starts have a pronounced 8 AM peak as well as a larger 5 PM peak; casual rides peak at 5 PM without a comparable morning surge.

6. Membership Marketing Campaign Recommendations

The findings suggest several opportunities for Cyclistic to test strategies that could encourage casual riders to become annual members.

1. Introduce weekend-focused membership campaigns

Cyclistic could introduce targeted weekend marketing campaigns highlighting the value of annual membership for casual riders who make frequent weekend trips. Weekend usage is a clear behavioural distinction between the two rider categories, providing a specific occasion around which to position membership.

- Highlight the value and potential cost savings of annual membership for riders who frequently use the service on weekends.
- Run membership campaigns during warmer months, when casual riding activity increases.
- Emphasise the convenience and flexibility of using Cyclistic bikes for weekend activities.

2. Introduce location-based membership promotions

Cyclistic could test location-based membership promotions around stations with high casual-rider activity. The geographical concentration of casual rides can help identify locations where membership promotions may reach a relevant audience.

- Display membership promotions through station signage or QR codes.
- Offer membership information at selected high-traffic casual-rider locations.
- Use digital advertising targeted at people visiting these areas.

3. Emphasise electric bike access in membership marketing

Cyclistic could emphasise the convenience and availability of electric bikes in its membership marketing, particularly when communicating with casual riders. Electric bikes are already popular among casual riders, providing an opportunity to communicate a feature that may appeal to this group.

- Highlight the availability of electric bikes to prospective members.
- Clearly communicate electric-bike benefits and any associated membership pricing or usage conditions.
- Test campaigns that emphasise the convenience of electric bikes for different types of journeys.

4. Use usage patterns to inform campaign timing

Cyclistic could use observed hourly and day-of-week usage patterns to inform the timing of membership campaigns. Different campaign timings could be tested to determine whether they improve engagement among casual riders.

These are potential strategies based on ride-level patterns. Campaign performance could be evaluated using engagement rates, membership sign-ups and conversion rates, allowing Cyclistic to compare the effectiveness of different campaigns.

Cyclistic should test these strategies in selected locations and during selected periods before rolling them out across the entire service area.

7. Limitations

- The dataset does not identify individual riders, so results describe trips rather than unique customers or their likelihood of converting to membership.
- Trips with missing station names were removed from all subsequent analyses, including overall ride counts and duration comparisons. If missingness differs by rider category, the reported comparisons may be affected.
- Nineteen duplicate ride_id entries were retained.
- Trip durations shorter than 60 seconds or longer than 24 hours were excluded; this may remove some unusual but legitimate trips.
- Station locations and hourly patterns can suggest possible trip purposes, but cannot confirm whether a ride was for commuting, recreation or tourism.
- The reporting period is historical and does not describe real-time usage.

8. Conclusion

The analysis identifies consistent differences between casual riders and annual members. Members account for more rides and show a stronger weekday pattern, while casual riders take longer rides and a greater proportion of their rides occur on weekends. Casual riders also have a higher electric-bike share and different concentrations of starting stations. These differences provide a basis for developing and testing targeted membership marketing, especially around weekend use and locations popular with casual riders. Any strategy should be validated through campaign testing and interpreted in light of the dataset's limitations.